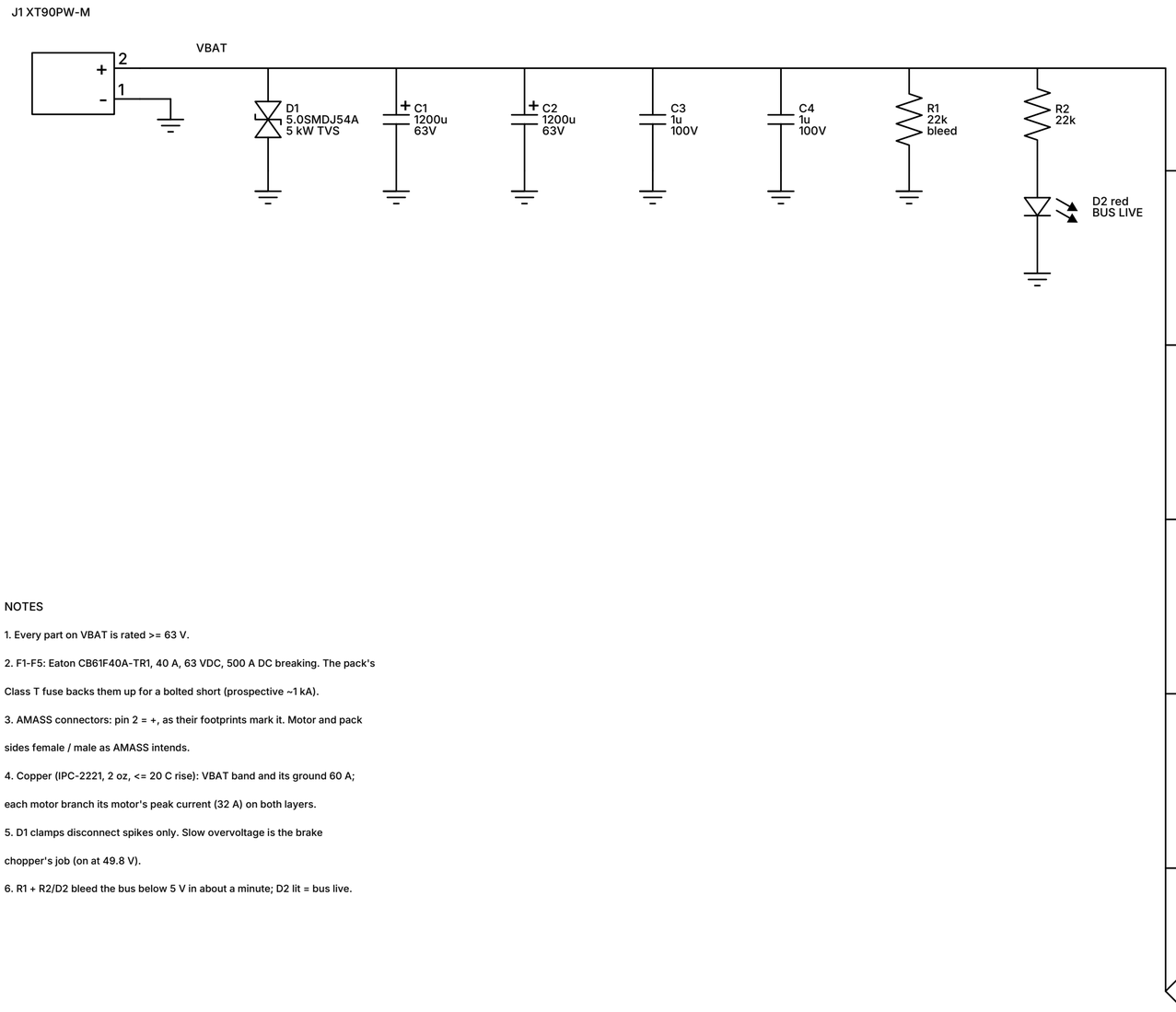
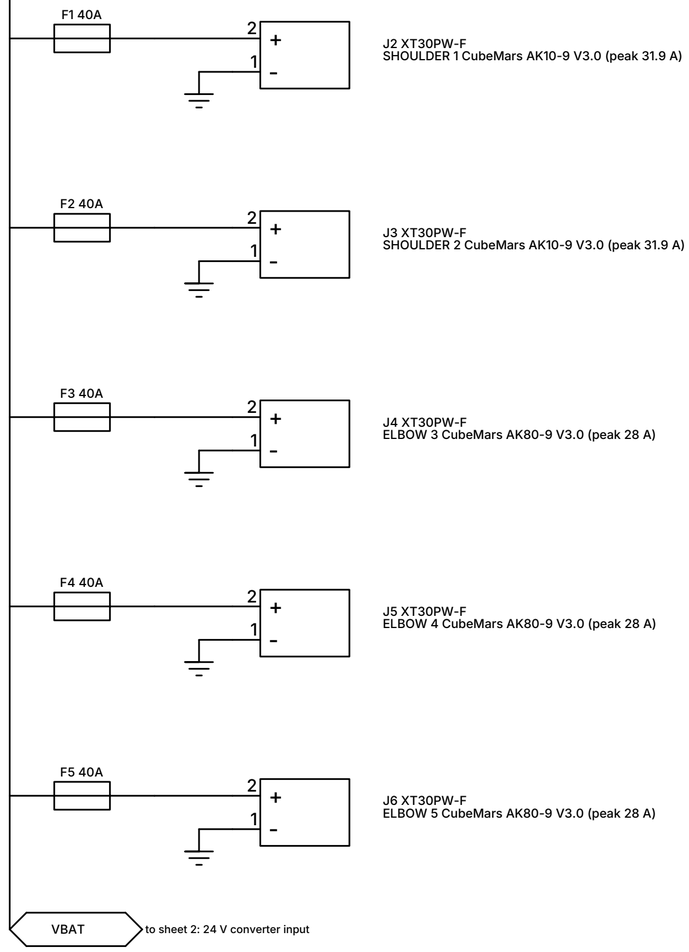
PACK INPUT - 12S Li-ion, 36.0-49.0 V (BMS OVP 50.4 V, drivers 52 V abs. max)



MOTOR FEEDS - 49 V, one 40 A fuse per motor



NOTES

1. Every part on VBAT is rated >= 63 V.

2. F1-F5: Eaton CB61F40A-TR1, 40 A, 63 VDC, 500 A DC breaking. The pack's Class T fuse backs them up for a bolted short (prospective ~1 kA).

3. AMASS connectors: pin 2 = +, as their footprints mark it. Motor and pack sides female / male as AMASS intends.

4. Copper (IPC-2221, 2 oz, <= 20 C rise): VBAT band and its ground 60 A; each motor branch its motor's peak current (32 A) on both layers.

5. D1 clamps disconnect spikes only. Slow overvoltage is the brake chopper's job (on at 49.8 V).

6. R1 + R2/D2 bleed the bus below 5 V in about a minute; D2 lit = bus live.

POWER DISTRIBUTION

PIONEER-ARM-PDB (cad_bench/projects/arm_pdb.py)

drawn by the harness; checked against the layout netlist

SHEET 1 / 2	REV A
DATE 2026-10-12	SIZE A3